THE CHINESE UNIVERSITY OF HONG KONG
Department of Statistics

STAT3003: Survey Methods
Beyond Formulas. - Exercises

1 The Two Major Categories of Survey Error

In any survey, the final estimate will likely differ from the true population value. These differences arise from two main types of error.

1.1 Errors of Non-observation (Data We Failed to Collect)

These errors occur when the final sample of respondents is not fully representative of the target population.

Sampling Error
- **Definition:** The inherent difference between an estimate derived from a sample and the true population parameter. It is an unavoidable consequence of sampling instead of conducting a census.
- **Reduction:** Can be reduced by using appropriate sampling designs (like stratification) and increasing the sample size. This is the type of error our formulas typically address.

Coverage Error
- **Definition:** Occurs when the sampling frame (the list from which the sample is drawn) does not perfectly cover the target population.
- **Example:** Using a phonebook as a sampling frame misses people with unlisted numbers or those without a landline.
- **Challenge:** This error is difficult to measure and correct.

Non-response Error
- **Definition:** A serious error that happens when data is not collected from some of the units selected for the sample.
- **Types:**
  1. *Inability to Contact:* The selected person cannot be reached (e.g., not at home).
  2. *Inability to Answer:* The person is unable to answer (e.g., doesn't know the information, language barrier).
  3. *Refusal to Answer:* The person is unwilling to participate, often due to privacy concerns or lack of time.
- **Key Concern:** Non-response introduces **bias** if the non-responding group differs systematically from the responding group (e.g., if only young people respond to a survey about technology use).

1.2 Errors of Observation (Data We Collected Incorrectly)

These errors occur during the data collection process itself, leading to inaccurate measurements even when a respondent is participating.

Interviewer Error
- **Definition:** The interviewer's presence or behaviour influences the respondent's answers.
- **Example (Schuman & Converse, 1971):** Black residents in Detroit were asked about trusting white people. 35% said they could trust "most" when the interviewer was white, but only 7% said the same when the interviewer was black.
- **Cause:** Intonation, emphasis, or perceived opinions of the interviewer can lead respondents to give socially desirable answers.

Respondent Error
- **Definition:** Errors originating from the respondent.

- **Types:**
    1. *Recall Bias:* Inaccurately remembering past events.
    2. *Prestige Bias:* Exaggerating answers to appear more favorable (e.g., overstating income).
    3. *Intentional Deception:* Deliberately providing false information.
    4. *Incorrect Measurement:* Misunderstanding the question or the units required.

Measurement Instrument Error
- **Definition:** The survey questions or questionnaire are poorly designed.
- **Cause:** Ambiguous wording, unclear definitions, or confusing layout.
- **Example:** Asking "How many glasses of water do you drink a day?" without defining the size of a 'glass'.

2 Practical Strategies for Reducing Errors

- **Callbacks:** Making multiple attempts to contact a respondent at different times of the day to reduce non-response.
- **Rewards and Incentives:** Offering a small monetary reward or token of appreciation to encourage participation. **Crucially**, this should be offered *after* a respondent is selected for the sample, not used to recruit volunteers (which would create a biased sample).
- **Interviewer Training:** Training interviewers to be neutral, read questions exactly as written, and build rapport without influencing answers.
- **Data Checks:** Performing logical checks on the collected data to find errors (e.g., a proportion cannot be greater than 1; an adult cannot be 5 years old).

3 The Art of Question Design

The wording and structure of a questionnaire are critical for collecting accurate data.

4 Likert Scales for Measuring Attitudes

- **Likert Item:** A statement followed by a symmetric response scale (e.g., Strongly Agree, Agree, Neutral, Disagree, Strongly Disagree).
- **Likert Scale:** The sum or average of scores from multiple Likert items. The goal is to measure a respondent's overall attitude towards a complex topic by combining their responses to several related statements.
- **Design Principles:**
    - Use an odd number of responses (typically 5 or 7) to provide a neutral midpoint.
    - Statements should be clear, simple, and address a single concept.
    - The collection of items should cover the key dimensions of the attitude being measured.

Table 1: Common Pitfalls in Question Design

Pitfall	Description	Solution / Best Practice
Question Order	The order in which questions are asked can influence responses. Answering a specific question can frame the respondent's mindset for a subsequent general question.	Ask general questions *before* specific ones. (e.g., Ask about general happiness before marital happiness).
Response Order	Respondents may favor choices at the beginning of a written list (**primacy effect**) or at the end of a spoken list (**recency effect**).	Randomize the order of responses for different subsets of the sample. Use showcards in personal interviews.
Closed vs. Open	**Closed:** Fixed choices. Easy to analyze but restrictive. **Open:** Free response. Provides rich detail but is difficult to quantify.	Use open questions in a pre-test to develop good response categories for a closed question in the main survey.
"Don't Know" Option	Forcing an opinion can lead to random answers. However, providing a "Don't Know" option can be an easy way out, reducing effective sample size.	Use **screening questions** first to gauge a respondent's knowledge. Only ask the question if they are knowledgeable enough.
Double-Barrelled	Asking about two different concepts in a single question.	Split the question into two separate questions. (e.g., "Do you agree with Bill Clinton's $50 billion bailout of Mexico?" mixes opinions on Clinton and the bailout).
Leading/Unbalanced	Phrasing that suggests a "correct" answer or presents only one side of an issue.	Balance the question. Instead of "Do you favor...?," ask "Do you favor or oppose...?" Avoid emotionally charged words ("forbid" vs. "not allow").

5 Interpreting Results with Caution

- **Confounding Variables:** A "third" variable that is associated with both the independent and dependent variables, creating a spurious association. **The main takeaway: association does not imply causation.**
- **Simpson's Paradox:** A trend that appears in different groups of data but disappears or reverses when these groups are combined.
  - **Classic Example: Berkeley Sex Bias (1973).**
    - *Aggregate Data:* Overall, men had a higher acceptance rate than women, suggesting bias against women.
    - *Disaggregated Data:* When looking at individual departments, most departments had similar acceptance rates for men and women. In fact, some departments showed a slight bias *in favor* of women.
    - *Explanation (Confounding):* The paradox was caused by a confounding variable: the **choice of department**. Women tended to apply to more competitive departments with lower overall acceptance rates, while men tended to apply to less competitive departments with higher acceptance rates.

6 The 11-Point Plan for a Successful Survey

A comprehensive plan is the best defense against the errors discussed.

1. **Statement of Objectives:** Clearly define what you want to measure.
2. **Target Population:** Precisely define the group you are studying.
3. **The Frame:** Identify the list(s) from which you will draw your sample.
4. **Sample Design:** Choose the appropriate method (SRS, stratified, etc.) and determine the sample size.
5. **Method of Measurement:** Decide how data will be collected (personal interview, phone, online).
6. **Measurement Instrument:** Design the questionnaire carefully.
7. **Selection and Training of Fieldworkers:** Prepare data collectors to perform their job consistently and without bias.
8. **The Pretest:** Conduct a small-scale trial survey to test your questionnaire and procedures.
9. **Organization of Fieldwork:** Plan the logistics of data collection.
10. **Organization of Data Management:** Plan how the collected data will be stored, cleaned, and processed.
11. **Data Analysis:** Decide *before* you start what statistical analyses you will perform to meet your objectives.

Exercise

1. (Adapted from Exercise 2.4 in Scheaffer et al. (2012)) Discuss problems associated with question ordering. Give a list of two or three questions for which you think order is important and explain why.

2. (Adapted from Exercise 2.9 in Scheaffer et al. (2012)) Discuss the use of open versus closed questions. Give an example of an appropriate open question. Give an example of how a similar question could be closed. What are the advantages of closed questions?

3. (Adapted from Exercise 2.15 in Scheaffer et al. (2012)) Why is the response rate an important consideration in surveys? Discuss methods for reducing the nonresponse rate.

4. (Adapted from Exercise 2.16 in Scheaffer et al. (2012)) Give an example of a question that contains a weak counterargument. Give an example of a question that contains as strong counterargument.

5. (Adapted from Exercise 2.19 in Scheaffer et al. (2012)) In a Gallup youth survey (*Gainesville Sun*, February 13, 1985), 414 high school juniors and seniors were asked the following question: What course or subject that you have studied in high school has been the best for preparing you for your future education or career? In their responses to this question, 25% of the students chose mathematics and 25% chose English. Do you think this is a good question with informative results?

6. (Adapted from Exercise 2.22 in Scheaffer et al. (2012)) In a study of HIV-related risk groups, as part of a telephone interview, people in the suvrey were asked intimate questions about their personal lives. Is there a possibility of bias in the responses? If so, in which direction? Will randomization in selecting the respondents help reduce potential bias related to the sensitive questions? Will randomization in selecting the respondents help reduce any potential bias?

THE END